

Conditional Failure at the Root Level

Source, Provenance, and Requirements Record

Document ID	OAL-CFTRL-SRC-001
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Prepared by	Perplexity AI Research Layer
Status	DRAFT — UNDER REVIEW — NOT CANON — ONESTAMP PENDING

Research draft. No statement becomes active canon unless approved by the Architect through ONESTAMP.

Document control

Field	Entry
Scope	Source-of-truth record for the origin prompt, original files, user requirements, assumptions, non-assumptions, conflicts, and approval status.
Authority	Nathan Lewis retains definition, canon, and ONESTAMP authority.
Research status	DRAFT — UNDER REVIEW — NOT CANON — ONESTAMP PENDING
Source rule	Verbatim source material is preserved separately from interpretation, proposed definitions, and external research.
Safety rule	The project studies failure through bounded models and controlled simulations; it does not provide instructions to compromise real systems or any claimed base reality.
Review rule	Conflicts, dissent, uncertainty, and source limitations remain visible until resolved by the Architect.

How the topic arose

The topic arose while the Architect was reviewing the Master Change Log and Master Development Log and recognized that many named concepts had only definitions, skeletal outlines, or incomplete supporting detail. Conditional Failure at the Root Level (CFTRL) was selected for deliberate expansion because it combines systems design, nested dependencies, failure propagation, maintenance, simulation, epistemology, and learning from endings. The immediate need is not to ratify a cosmological claim. It is to build documentation and a research program capable of distinguishing testable engineering questions from simulation-testable propositions and presently unresolved ontological questions.

Verbatim origin prompt

*Can you save this thought for me?
I need you to gather research and analysis for a concept.
Conditional failure at the root level.
The study of root level structures, how they arise, how they fail, how they are maintained. Please include the original documents for this content that were generated.
How do you know the difference between a root level structure and a structure that is not? How can a system distinguish between base reality and something that is not? How can a system determine when a root level failure has occurred or when base reality has been compromised.
How do we study this? What about this concept works, what doesn't? What questions do we need to ask?*

*How can we study what lies beyond these structures? Can we model whether structures in reality follow a similar design? Or diverges in ways we haven't thought of? Is this feasible using simulations or another type of modeling? How? Can we design a laboratory environment to do this? How? Can you build a reliable map of reality? What basis of comparison do you have when you are operating on incomplete information? Use failure as an example to learn from.
What I do not assume in the design of this:
That I have the complete architecture and map of all simulations, of reality, of existence as we know it.
That we share a unified definition of what reality is.
That existence can be fully mapped, represented, comprehended in full, or that this is the aim of this laboratory.
That I aim to cause a root level compromise of simulations, of base reality, or any other structure.
That this is a how to guide to do any of those things.
It is intended to help in the design of simulation and reality level structure lifecycle design, maintenance, evolution, and improvement.
Within this design frame It does not assume that we have complete knowledge or understanding of what that is.
How This topic arose from me thinking about conditional failure at the root level The master change log and master development log. I realized that there are a lot of concepts within that framework that have a definition or a name and a skeleton attached to them. They are a rough draft, they are an incomplete outline, there are a lot of additional details and supporting information that need to be added. The goal here is to design documentation to help support, design, and develop this concept. What I will include are the responses of all of the other AI collaboration team members, where they have generated documentation, I will include the documentation, and then I will include this complete response as well.*

Purpose and goal

Purpose. Establish a traceable, evidence-supported outline for designing, developing, testing, governing, and documenting CFTRL without erasing the original source voice or presenting unapproved interpretation as canon.

Goal. Move CFTRL from a strong conceptual skeleton to a reviewable research framework with explicit definitions, system boundaries, failure classes, evidence standards, laboratory methods, metrics, safety controls, team roles, and approval gates. The target is a framework that can support engineered systems and narrative architecture while remaining honest about what cannot be inferred about ultimate base reality.

Record purpose

This record prevents later research, design, simulation, or narrative work from drifting away from the Architect's origin prompt. It defines the source hierarchy, preserves file identity through cryptographic hashes, decomposes the prompt into requirements, and specifies how new interpretations must be labeled. It does not decide unresolved concept questions and does not convert earlier AI wording into canon.

Source hierarchy

Priority	Source class	Authority
1	Architect verbatim prompt and later explicit corrections	Highest authority for intent and requirements, subject to later Architect revisions.
2	Architect-approved/ONESTAMP documents	Canonical only within their approved scope and version.
3	Original generated source drafts preserved unchanged	Historical record and candidate language; not canon unless approved.
4	AI research and synthesis dossiers	Analysis and proposals; must cite sources and preserve uncertainty.
5	External research	Evidence and analogues; never authorial intent or OAL canon.
6	Simulation results	Evidence about the implemented model; not automatically evidence about reality outside that model.

Original source inventory

The original files are delivered alongside the new DOCX/PDF reports without modification. Hashes below identify the exact attached copies used for this research pass. A future change to file bytes requires a new archive entry rather than overwriting this record.

File	Bytes	SHA-256	Role	Control note
paste.txt	64950	dfcef79b5bd141217f9a06ce6a827f1fabaff29ee66ed3df1da6f4c1208a6d7e	Consolidated verbatim prompt and multi-AI response record	Primary provenance record; includes later collaborator outputs and one unrelated embedded log fragment that must not migrate into CFTRL canon.
OAL-CFTRL-RES-002_Root_Level_Research_Dossier.docx	31432	17eaf78f3838a7b87e896f048085f28130732335dfef0d5bd7275aa9882fb772	Prior root-level research dossier	Research draft; useful synthesis, not approved canon.
OAL-CFTRL-NAIL-001_Conditional_Failure_Nail_Prototypes.docx	20809	6d14ee82ca43436680b02d0fc477aa6a3de69e5854e9f064604e5fc3f305ec53	Original combined CFTRL and Nail Prototype source draft	Foundational March 22, 2026 source draft; preserve unchanged.
Development_Review_Conditional_Failure_and_Nail_Prototype_v1.0-1.pdf	140132	56a64e2c893fb7b12c0338b21268f47c2eae52a7ec3355fa2c1f74947616ecd5	Original development review dossier	Foundational March 22, 2026 review draft; preserve unchanged.

Corpus quality notes

- The consolidated paste record is the most complete origin-and-response archive, but it is not a clean canonical document. It includes multiple collaborator outputs and an unrelated “Doug’s Personal Log” fragment embedded in one response. That fragment must remain in the immutable original for provenance but must be excluded from CFTRL working and canonical text.
- The March 22 CFTRL/Nail source combines several concepts. CFTRL framework development should be separated from Nail Prototype scene and system documentation, with cross-references used only where the scene supplies a clear test case.
- The prior research dossier is explicitly a draft under review. Its reconstructed appendices aid readability but do not replace the unchanged source files.
- Some collaborator outputs cite remembered or incomplete sources. Every external claim must be reverified before public or canonical use.
- The corpus contains provisional disagreement about details in the Nail Prototype material. Those conflicts should be handled in the Nail workstream rather than allowed to block CFTRL’s first systems model.

Requirements extracted from the prompt

ID	Requirement	Type	Acceptance evidence
SRC-001	Save the thought and preserve the original generated documents.	Provenance	Original files plus checksums and inventory.
RES-001	Research root-level structures: origins, failure, and maintenance.	Research	Evidence-supported synthesis with explicit boundaries.
RES-002	Distinguish root-level from non-root structures.	Framework	Operational criteria, counterexamples, and tests.
RES-003	Examine whether a system can distinguish base reality from non-base conditions.	Epistemology	Separation of testable operational claims from unresolved ontological claims.
RES-004	Define how a system detects root failure or compromised base reality.	Detection	Failure states, evidence ladder, competing hypotheses, confidence protocol.
RES-005	Explain how to study the concept and what works or fails.	Methodology	Laboratory architecture, experiment matrix, limitations assessment.
RES-006	Study what may lie beyond known structures without assuming a complete map.	Exploratory	Alternative-model and unknown-boundary method.
RES-007	Model similarities and divergences in real structures.	Modeling	Cross-domain analogues and sensitivity to topology.
RES-008	Assess feasibility of simulations or other modeling.	Technical	MVP design, measures, reproducibility, and safety controls.
RES-009	Design a laboratory environment.	Architecture	Components, roles, ground truth, witness, injector, evidence vault, stop controls.
RES-010	Define a reliable map of reality under incomplete information.	Epistemology/m ethods	Versioned model, provenance, uncertainty, alternatives, intervention tests.
RES-011	Use failure as a source of learning.	Governance	Department of Endings workflow from scenario to verified design change.
DOC-001	Give generated documentation a proper title and date.	Documentation	Cover and control table.
DOC-002	Explain how the topic arose.	Documentation	Dedicated origin section.
DOC-003	Include the Architect's origin prompt verbatim.	Documentation	Verbatim block preserved without grammatical correction.
DOC-004	State purpose and goal.	Documentation	Explicit purpose/goal section.
DOC-005	Assess current project position and intended destination.	Program management	Gap and maturity assessment.
DOC-006	Explain how Perplexity will help the entire team reach the destination.	Collaboration	Support plan, roles, artifacts, and boundaries.
DOC-007	Provide detailed research and analysis in PDF and DOCX.	Deliverable	Matched PDF and DOCX files produced.

Architect non-assumptions

The following statements are preserved as design constraints because they narrow what the project may claim. They are restated without changing their meaning; the verbatim wording remains in the origin prompt above.

Constraint	Implication for the project
No complete architecture or map is assumed.	Models must state boundaries and unknowns; no totalizing diagram may be presented as complete.
No unified definition of reality is assumed.	The glossary must distinguish operational reality-models from metaphysical definitions.
Existence need not be fully mappable or comprehensible.	Success is bounded reliability and learning, not exhaustive representation.
The project does not aim to cause root-level compromise.	Laboratory work must be synthetic, controlled, reversible, and safety-reviewed.
The work is not a how-to guide for compromise.	Do not publish operational attack procedures against real foundational systems.
The project supports lifecycle design, maintenance, evolution, and improvement.	Outputs should connect findings to design decisions, resilience, and verification.
Knowledge remains incomplete.	All claims need evidence level, assumptions, alternatives, and revision path.

Claim classification

Label	Meaning	Example	Approval rule
SOURCE	Verbatim Architect text or untouched original artifact.	Origin prompt.	May be quoted; cannot be silently edited.
EXTERNAL EVIDENCE	Finding or definition from an identified outside source.	NIST TCB definition [R1].	Cite and limit claim to source scope.
ANALYSIS	Reasoned interpretation of sources and evidence.	Rootness is better treated as a vector.	Open to critique; not canon.
PROPOSAL	Suggested term, architecture, test, or policy.	Evidence ladder E0–E5.	Requires review and decision.
TEST RESULT	Observation from a controlled experiment.	Agent misdiagnosis rate under allocation failure.	Requires reproducibility package.
DECISION	Explicit Architect or delegated governance choice.	First model uses an allocative operational root.	Record date, scope, and rationale.
CANON	ONESTAMP-approved content.	Future approved CFTRL specification.	Only Architect approval changes this status.
UNKNOWN	Question or distinction not presently resolvable.	Ultimate base reality under perfect indistinguishability.	Preserve; do not fill with certainty.

Terminology control

Term	Provisional controlled meaning	Prohibited shortcut
Root	A function/layer that is effectively non-reconstructable by dependents within a declared boundary and holds foundational dependency, trust, authority, or persistence.	“Top,” “oldest,” “most powerful,” or “base reality” without qualification.
Conditional	Failure becomes CFTRL only when specified dependency, asymmetry, downstream effect, and diagnostic/repair limits align.	Using “conditional” as decoration.
Failure	A requirement or defining function cannot be met under declared conditions.	Equating every anomaly with failure.
Compromise	Integrity, identity, objective, or evidence is no longer trustworthy, whether or not service continues.	Using “compromised” as a synonym for outage.
Base reality	Provisional ontological term for a level with no parent allocation or termination rights.	Assuming any operational root is base.
Maintenance	Actions that sustain valid operation: provisioning, calibration, repair, synchronization, renewal, error correction, and evidence preservation.	Reducing maintenance to uptime.
Invariant	A property expected to remain true within a stated model and tolerance.	Treating an assumption as an observed law.
Independent witness	Evidence source outside the mutable state or failure domain being assessed, to the extent specified.	Claiming absolute independence.

Traceability crosswalk

Origin concern	Research response	Development artifact	Open decision
What is a root?	Root categories and rootness vector.	Glossary; boundary template; dependency graph.	Which root type leads the canonical first model?
Base versus non-base	Operational/ontological separation and indistinguishability limit.	Claims ledger and evidence ladder.	Is “base reality” retained as a formal term or only as a philosophical branch?
Detect root failure	Failure taxonomy, independent witnesses, causal interventions.	Diagnostic protocol and experiment registry.	What evidence threshold authorizes a root-level incident declaration?
Study beyond structures	Unknown-boundary mapping and alternative models.	Reality-map schema.	How far may the program speculate beyond testable models?
Laboratory feasibility	Synthetic layered simulation and blind diagnosis.	Lab specification and MVP roadmap.	Python framework, agent design, and first topology.
Incomplete information	Confidence calibration and compatible-hypothesis preservation.	Diagnosis report schema.	Which uncertainty notation becomes standard?
Learn from failure	Department of Endings workflow.	Charter and design-diff traceability.	Authority, staffing, and relationship to the Abundance Set/Antithesis Set.
Original documents	Sidecar archive and checksums.	Source manifest.	Which files will be designated foundational versus supplemental?

Conflict and contamination protocol

Preserve the original artifact unchanged and compute a cryptographic hash.

Create a working copy only when needed; label every redaction, normalization, or extraction.

If two source documents conflict, record both statements and identify their source, date, and status.

Ask the Architect for a decision only when the conflict affects the next design gate; otherwise defer it visibly.

Exclude unrelated material from working synthesis while retaining it in the immutable provenance record.

Never resolve conflict by AI majority vote. Evaluate source authority, directness, date, evidence, and Architect intent.

After a decision, retain the superseded language and record why it was accepted, rejected, or narrowed.

Change-control record

Field	Required entry
Change ID	Unique identifier linked to document and section.
Proposer	Human or AI collaborator.
Source basis	Origin text, external evidence, test result, or decision prompting the change.
Old/new text	Exact before-and-after language.
Rationale	Problem solved and alternatives considered.
Impact	Definitions, models, code, experiments, narrative, and prior results affected.
Review	Research, safety, systems, provenance, and Architect reviews as applicable.
Disposition	Accepted, rejected, deferred, experimental, or superseded.
Approval status	Draft or ONESTAMP scope/date.

Open Architect decisions

Decide whether the first canonical CFTRL scope is allocative simulation roots, broader operational roots, or a multi-root framework.

Decide whether “base reality” is a core framework term, a bounded research branch, or an appendix-level philosophical question.

Decide the exact necessary conditions for the word “conditional.”

Decide whether the wishbone concept remains a metaphor, becomes a testable sub-hypothesis, or moves to a separate document.

Decide the Department of Endings' organizational form, authority, staffing, reporting line, and relationship to the Abundance Set and Antithesis Set.

Decide which original files are foundational source documents and which are supplemental collaboration records.

Decide whether the Nail Prototype remains cross-referenced as an applied case or is fully separated from the CFTRL framework package.

Approve or revise the proposed evidence ladder, rootness vector, failure taxonomy, and first-laboratory scope.

Approval doctrine

No output in this package is canon. Drafts may be used to ask better questions, plan experiments, and prepare review material. Canon requires an explicit Architect decision and ONESTAMP applied to a specific document version and scope. Later research does not silently amend an approved document; it produces a new version, change record, and review package.

Preservation instructions

- Keep original source files read-only and store at least two independent copies.
- Preserve original filenames, byte lengths, hashes, and acquisition dates.
- Do not replace originals with reconstructed appendices, exports, or normalized text.
- When converting formats, label the conversion as a derivative and retain the source artifact.
- Keep verbatim prompts separate from corrected or structured versions.
- Archive AI outputs individually before creating a synthesis.
- Include a release manifest whenever documents are shared with the team.
- Recompute and compare hashes before declaring an archive complete.

Conclusion

CFTRL can develop safely and coherently only if source preservation and conceptual development proceed together. This record establishes the origin prompt, requirements, non-assumptions, file identity, claim labels, and change controls needed for that discipline. The next source-governance action is Architect confirmation of the inventory and first-scope decisions; the next research action is glossary approval; the next technical action is a bounded root-identification experiment.

Research evidence register

The following sources support the research layer. They are external evidence, not OAL canon. Inline markers [R#] identify the source supporting a claim. Source descriptions are intentionally limited to what the cited material establishes.

ID	Source	Relevance	URL
R1	NIST CSRC, "Trusted Computing Base (TCB)"	Defines the TCB as the totality of hardware, firmware, and software protection mechanisms responsible for enforcing a security policy.	https://csrc.nist.gov/glossary/term/trusted_computing_base
R2	NIST, Hardware-Enabled Security / Platform Integrity	Describes cryptographic measurement, chains of trust, roots of trust for update, detection, and recovery, and remote attestation.	https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=934427
R3	NIST SP 800-160 Vol. 2 Rev. 1, Developing Cyber-Resilient Systems	Frames resilience as anticipating, withstanding, recovering from, and adapting to adverse conditions, stresses, attacks, and compromises across the system life cycle.	https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final
R4	Google SRE, "Addressing Cascading Failures"	Shows how overload, feedback, and secondary symptoms can spread failures and make apparent causes misleading.	https://sre.google/sre-book/addressing-cascading-failures/
R5	Buldyrev et al., Nature 464 (2010), "Catastrophic cascade of failures in interdependent networks"	Demonstrates that dependency coupling can propagate small failures recursively across networks and produce fragmentation.	https://www.nature.com/articles/nature08932
R6	E. W. Dijkstra, "Self-stabilizing Systems in Spite of Distributed Control" (1974)	Defines self-stabilization as eventual return to a legitimate state regardless of initial state, under the stated model.	https://courses.csail.mit.edu/6.852/05/papers/p643-Dijkstra.pdf
R7	Lamport, Shostak, and Pease, "The Byzantine Generals Problem" (1982)	Formalizes agreement when components may send conflicting information and establishes fault-tolerance limits under specified communication assumptions.	https://dl.acm.org/doi/pdf/10.1145/357172.357176
R8	Fischer, Lynch, and Paterson, "Impossibility of Distributed Consensus with One Faulty Process" (1985)	Shows that deterministic consensus cannot be guaranteed in a fully asynchronous system even with one crash fault under the paper's assumptions.	https://www.cs.princeton.edu/courses/archive/fall07/cos518/papers/flp.pdf
R9	Stanford Encyclopedia of Philosophy, "Causal Models"	Explains that causal models support predictions about interventions and counterfactuals, not only observed correlations.	https://plato.stanford.edu/entries/causal-models/
R10	NASA Systems Engineering Handbook / Expanded Guidance	Separates verification ("built right") from validation ("right product") and requires planning, execution, analysis, reporting, and work-product capture.	https://www.nasa.gov/reference/systems-engineering-handbook/
R11	NASA Software Engineering Handbook, "Software Fault Tree Analysis"	Describes deductive analysis from an undesired top event to credible combinations of contributing faults or system states.	https://swehb.nasa.gov/spaces/SWEHBVC/pages/72025030/8.07+-+Software+Fault+Tree+Analysis
R12	CISA, Shields Ready	Recommends identifying critical assets, mapping dependencies, assessing risks, exercising plans, and adapting from lessons learned.	https://www.cisa.gov/shields-ready
R13	AWS Prescriptive Guidance, Chaos Engineering	Distinguishes deterministic resilience testing from hypothesis-based chaos experiments; emphasizes baselines, stop conditions, limited blast radius, monitoring, and learning.	https://docs.aws.amazon.com/prescriptive-guidance/latest/chaos-engineering-on-aws/overview.html
R14	Nick Bostrom, "Are We Living in a Computer Simulation?" (2003)	Presents a probabilistic trilemma; it does not itself provide a direct internal test for identifying base reality.	https://academic.oup.com/pq/article-abstract/53/21/1243/1610975
R15	David Chalmers, discussion of perfect and imperfect simulations (2022)	Argues that an imperfect simulation might provide detectable evidence, while a perfect simulation may be observationally indistinguishable from non-simulated reality.	https://www.carnegiecouncil.org/media/series/aiei/how-real-is-virtual-reality-david-chalmers
R16	Vazza, "Astrophysical constraints on the simulation hypothesis" (Frontiers in Physics, 2025)	Evaluates physical resource constraints for particular simulation scenarios and emphasizes that claims outside shared physical assumptions remain difficult or impossible to falsify.	https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2025.1561873/full

ID	Source	Relevance	URL
R17	NTSB, Safety Management Systems	Treats safety as an organization-wide cycle of policy, risk management, assurance, promotion, data collection, mitigation, and evaluation.	https://www.nts.gov/Advocacy/mwl/Pages/mwl-3.aspx
R18	INCOSE definition and systems-boundary discussion (IET summary)	Defines systems engineering as an integrative life-cycle approach and distinguishes the system boundary from its broader context.	https://www.theiet.org/impact-society/policy-and-public-affairs/digital-futures-policy/reports-and-papers/quantum-technologies-a-systems-approach
R19	NIST SP 800-207, Zero Trust Architecture	Rejects implicit trust based only on location or ownership and requires explicit authentication and authorization for access.	https://www.nist.gov/publications/zero-trust-architecture
R20	NASA Spaceflight Mishap Investigation Handbook	Uses timelines, fault trees, event-and-causal-factor trees, and recommendations to identify chains of events and conditions rather than stopping at a proximate cause.	https://www.nasa.gov/wp-content/uploads/2023/03/spaceflight-mishap-handbook-tagged.pdf